

Flightradar24 Setup Guide

Raspberry Pi Model B Plus — Raspbian Bookworm

RTL-SDR / Edimax EW-7811Un WiFi Dongle / FR24 Feed

1. Overview

This guide describes the complete setup of a Flightradar24 ADS-B feeder on a Raspberry Pi Model B Plus (ARMv6/BCM2835) running Raspbian Bookworm. It covers WiFi configuration, fr24feed and dump1090 installation, the local web map interface, persistent logging, WiFi watchdog, and WiFi driver recovery.

Architecture:

- dump1090-mutability — reads from RTL-SDR dongle, writes JSON, serves Beast TCP on port 30005
- fr24feed — connects to dump1090 via Beast TCP on port 30005, feeds data to Flightradar24
- lighttpd — serves the local gmap.html flight map web interface
- wifi-watchdog — cron script monitoring WiFi connectivity every 5 minutes
- wlan-driver-check — systemd service recovering WiFi driver if missing at boot

 *The Raspberry Pi Model B Plus uses ARMv6. Only fr24feed version 1.0.55-1 or later supports ARMv6. Always pin the package after install to prevent auto-upgrades breaking it.*

2. Prerequisites

- Raspberry Pi Model B Plus with fresh Raspbian Bookworm SD card
 - Edimax EW-7811Un WiFi USB dongle (RTL8188CUS chip)
 - RTL-SDR USB dongle with ADS-B antenna
 - Temporary wired ethernet connection for initial setup
 - Your Flightradar24 sharing key (fr24key)
 - Your WiFi SSID and password
-

3. WiFi Configuration

The Edimax EW-7811Un dongle uses the RTL8188CUS chip which has issues with WPA3 and power management on Bookworm. Follow these steps carefully.

3.1 Verify the dongle is detected

```
lsusb
```

You should see: Edimax Technology Co., Ltd EW-7811Un 802.11n Wireless Adapter [Realtek RTL8188CUS]

3.2 Scan for available networks

```
sudo nmcli device wifi rescan && sleep 3 && sudo nmcli device wifi list
```

Your network should appear in the list. If it does not, run the rescan command again.

3.3 Create a WPA2-forced connection profile

The RTL8188CUS cannot handle WPA3 negotiation. Force WPA2 explicitly:

```
sudo nmcli connection add type wifi ifname wlan0 con-name "YourSSID" ssid
"YourSSID" \
  wifi-sec.key-mgmt wpa-psk \
  wifi-sec.psk "YourPassword" \
  wifi-sec.proto rsn \
  wifi-sec.pairwise ccmp \
  wifi-sec.group ccmp
```

```
sudo nmcli connection up "YourSSID"
```

⚠ If connection fails with 'network not found', try: `sudo nmcli radio wifi off && sudo nmcli radio wifi on` — then retry the connection up command.

3.4 Disable WiFi power saving

Without this, the WiFi connection will drop after a few minutes and never reconnect:

```
sudo nano /etc/NetworkManager/conf.d/wifi-powersave-off.conf
```

Add the following content:

```
[connection]
wifi.powersave = 2
```

```
sudo systemctl restart NetworkManager
```

3.5 Verify WiFi is working

```
ip a s wlan0
ping -c 4 8.8.8.8 -I wlan0
```

wlan0 should have an IP address and ping should show 0% packet loss.

4. Blacklist RTL-SDR Kernel Modules

The default DVB kernel modules will grab the RTL-SDR dongle and prevent fr24feed from using it. Blacklist them permanently:

```
sudo nano /etc/modprobe.d/blacklist-rtl.conf
```

Add the following lines:

```
blacklist dvb_usb_rtl28xxu
blacklist rtl2832
blacklist rtl2830
```

```
blacklist e4000
```

💡 *The WiFi dongle uses rtl8192cu which is a different driver — do NOT blacklist it.*

```
sudo reboot
```

Verify after reboot that the modules are not loaded:

```
lsmod | grep -E 'rtl2832|dvb_usb_rtl'
```

This should return no output.

5. Install fr24feed

5.1 Add the FR24 repository

⚠️ *Use HTTP (not HTTPS) — the FR24 repo has a TLS certificate mismatch that blocks HTTPS.*

```
sudo bash -c "echo 'deb [trusted=yes] http://repo.feed.flightradar24.com  
flightradar24 raspberrypi-stable' > /etc/apt/sources.list.d/fr24feed.list"
```

```
sudo apt-get update
```

5.2 Install fr24feed

```
sudo apt-get install fr24feed
```

5.3 Pin fr24feed immediately

This prevents auto-upgrades from replacing the ARMv6-compatible binary with an incompatible one:

```
sudo apt-mark hold fr24feed
```

5.4 Verify the binary works

```
fr24feed --version
```

Should output: 1.0.55-1 (or later). If it shows 'Segmentation fault', the wrong architecture package was installed.

6. Install dump1090-mutability

6.1 Install the package

```
sudo apt-get install dump1090-mutability
```

6.2 Create the symlink for fr24feed

fr24feed looks for dump1090 at /usr/lib/fr24/dump1090 but it is installed at /usr/bin/dump1090-mutability:

```
sudo ln -s /usr/bin/dump1090-mutability /usr/lib/fr24/dump1090
```

6.3 Enable and configure dump1090

```
sudo sed -i 's/START_DUMP1090="no"/START_DUMP1090="yes"/'
/etc/default/dump1090-mutability
sudo sed -i 's/NET_BIND_ADDRESS="127.0.0.1"/NET_BIND_ADDRESS="0.0.0.0"/'
/etc/default/dump1090-mutability
```

Verify the changes:

```
grep START_DUMP1090 /etc/default/dump1090-mutability
grep NET_BIND_ADDRESS /etc/default/dump1090-mutability
```

6.4 Enable and start dump1090

```
sudo systemctl enable dump1090-mutability
sudo systemctl start dump1090-mutability
sudo systemctl status dump1090-mutability
```

Verify it is listening on the expected ports:

```
sudo ss -tlnp | grep dump1090
```

You should see ports 30001, 30002, 30003, 30004, 30005, 30104 all listening.

7. Configure fr24feed

7.1 Create the configuration file

```
sudo nano /etc/fr24feed.ini
```

Add the following content (replace YOUR_FR24KEY with your actual key):

```
receiver="beast-tcp"
host="127.0.0.1:30005"
fr24key="YOUR_FR24KEY"
bs="no"
raw="no"
logmode="0"
logpath="/var/log/fr24feed"
mlat="no"
mlat-without-gps="no"
bind-interface="0.0.0.0"
```

 **The receiver=beast-tcp mode makes fr24feed connect to the already-running dump1090 instance rather than spawning its own. This is required for the local web map to work correctly.**

7.2 Enable and start fr24feed

```
sudo systemctl enable fr24feed
sudo systemctl start fr24feed
sudo systemctl status fr24feed
```

Verify fr24feed is receiving and sending aircraft data:

```
sudo journalctl -u fr24feed -n 20 --no-pager
```

Look for lines like: [feed][i]sent 35,0 AC — this confirms data is flowing.

8. Install the Web Map Interface

8.1 Install lighttpd

```
sudo apt-get install lighttpd
```

8.2 Enable the dump1090 module

```
sudo lighty-enable-mod dump1090
sudo service lighttpd force-reload
```

8.3 Verify the web interface is working

Open a browser and go to:

```
http://<your-pi-ip>/dump1090/gmap.html
```

Aircraft should be visible on the map with no error message.

💡 Find your Pi's IP address with: `ip a s wlan0`

9. Persistent Journaling

By default, logs are lost on reboot. Enable persistent storage to allow post-mortem investigation of crashes or network drops.

9.1 Create the journal directory and enable persistence

```
sudo mkdir -p /var/log/journal
sudo systemd-tmpfiles --create --prefix /var/log/journal
```

9.2 Configure journald

```
sudo nano /etc/systemd/journald.conf
```

Set or uncomment the following values:

```
Storage=persistent
SystemMaxUse=200M
SystemKeepFree=500M
SystemMaxFileSize=50M
MaxRetentionSec=1month
```

```
sudo systemctl restart systemd-journald
```

Verify disk usage and available boots:

```
journalctl --disk-usage
journalctl --list-boots
```

9.3 Investigating previous boots

After any unexplained reboot or network outage, use these commands:

```
journalctl --list-boots
journalctl -b -1 -u NetworkManager
journalctl -b -1 -k | grep -iE 'wlan|rtl|disconnect'
journalctl -b -1 | grep wifi-watchdog
journalctl -b -1 | grep wlan-driver-check
```

10. WiFi Watchdog & Nightly Reboot

The WiFi watchdog runs every 5 minutes and automatically reconnects WiFi if it drops. If reconnection fails, it reboots the Pi. A nightly reboot at 3 AM serves as a final safety net.

10.1 Create the watchdog script

```
sudo nano /usr/local/bin/wifi-watchdog.sh
```

Add the following content:

```
#!/bin/bash
IFACE="wlan0"
PING_TARGET="8.8.8.8"

if ! ping -c 2 -I $IFACE $PING_TARGET &>/dev/null; then
    logger "wifi-watchdog: wlan0 unreachable, attempting reconnect..."
    nmcli device disconnect $IFACE
    sleep 5
    nmcli device connect $IFACE
    sleep 10
    if ! ping -c 2 -I $IFACE $PING_TARGET &>/dev/null; then
        logger "wifi-watchdog: reconnect failed, rebooting..."
        /sbin/shutdown -r now
    else
        logger "wifi-watchdog: reconnect successful"
    fi
fi
```

```
sudo chmod +x /usr/local/bin/wifi-watchdog.sh
```

Test it manually (should run silently if WiFi is working):

```
sudo /usr/local/bin/wifi-watchdog.sh
journalctl | grep wifi-watchdog
```

10.2 Add to crontab

```
sudo crontab -e
```

Add these two lines:

```
*/5 * * * * /usr/local/bin/wifi-watchdog.sh
0 3 * * * /sbin/shutdown -r now
```

 *The watchdog logs to the system journal. After any outage check: `journalctl -b -1 | grep wifi-watchdog`*

11. WiFi Driver Recovery at Boot

The rtl8192cu WiFi driver can occasionally fail to initialize at boot (especially after kernel updates), leaving wlan0 completely absent despite the dongle being detected at USB level. This service detects and recovers from that condition automatically.

11.1 Create the driver check script

```
sudo nano /usr/local/bin/wlan-driver-check.sh
```

Add the following content:

```
#!/bin/bash
# Wait for system to settle
sleep 30

# Check if wlan0 interface exists
if ! ip link show wlan0 &>/dev/null; then
    logger "wlan-driver-check: wlan0 not found, reloading rtl8192cu driver..."
    modprobe -r rtl8192cu rtl_usb rtl8192c_common rtlwifi 2>/dev/null
    sleep 2
    modprobe rtl8192cu
    sleep 10
    if ip link show wlan0 &>/dev/null; then
        logger "wlan-driver-check: wlan0 now available, bringing up connection..."
        nmcli connection up "Malson"
        logger "wlan-driver-check: connection up result: $?"
    else
        logger "wlan-driver-check: wlan0 still not available after driver reload!"
    fi
else
    logger "wlan-driver-check: wlan0 OK"
fi
```

 **Replace "Ma1son" with your actual WiFi connection name as shown in: nmcli connection show**

```
sudo chmod +x /usr/local/bin/wlan-driver-check.sh
```

11.2 Create the systemd service

```
sudo nano /etc/systemd/system/wlan-driver-check.service
```

Add the following content:

```
[Unit]
Description=Check and recover WiFi driver at boot
After=network.target NetworkManager.service
Wants=NetworkManager.service

[Service]
Type=oneshot
ExecStart=/usr/local/bin/wlan-driver-check.sh
RemainAfterExit=yes
SuccessExitStatus=0 4

[Install]
```

```
WantedBy=multi-user.target
```

11.3 Enable and start the service

```
sudo systemctl daemon-reload
sudo systemctl enable wlan-driver-check.service
sudo systemctl start wlan-driver-check.service
sudo systemctl status wlan-driver-check.service
```

After reboot, verify it ran correctly:

```
journalctl -b 0 | grep wlan-driver-check
```

You should see either 'wlan0 OK' (driver loaded normally) or the recovery sequence ending with 'connection up result: 0' (driver was reloaded and WiFi reconnected).

12. Final Verification

After a reboot, verify all services started correctly:

```
sudo reboot
```

After reboot:

```
sudo systemctl status fr24feed
sudo systemctl status dump1090-mutability
sudo systemctl status lighttpd
sudo systemctl status wlan-driver-check
ps aux | grep dump1090
ip a s wlan0
ping -c 4 8.8.8.8 -I wlan0
```

Expected results:

- fr24feed: active (running)
- dump1090-mutability: active (running)
- lighttpd: active (running)
- wlan-driver-check: active (exited) — with 'wlan0 OK' or successful recovery in journal
- Only ONE dump1090 process visible in ps aux
- wlan0 has an IP address
- Ping shows 0% packet loss

Also check the FR24 dashboard and local map:

- <http://<your-pi-ip>:8754/> — FR24 feeder dashboard showing aircraft count
 - <http://<your-pi-ip>/dump1090/gmap.html> — local map with no error message
-

13. Known Issues & Solutions

fr24feed segfaults on --version

Cause: Wrong architecture package (armhf) installed on ARMv6 Pi.

Solution: Ensure fr24feed version 1.0.55-1 or later is installed. Contact FR24 support if the

repo only provides 1.0.55-0.

WiFi connects but drops after ~10 minutes

Cause: WiFi power saving mode enabled.

Solution: Create `/etc/NetworkManager/conf.d/wifi-powersave-off.conf` with `wifi.powersave = 2` (see section 3.4).

WiFi shows 'network not found' despite network being visible in scan

Cause: WPA3 negotiation failure with RTL8188CUS chip, or driver state issue.

Solution: Force WPA2 using the nmcli connection profile (see section 3.3), or toggle WiFi radio: `sudo nmcli radio wifi off && sudo nmcli radio wifi on`

wlan0 missing after boot (driver not loaded)

Cause: `rtl8192cu` driver failed to initialize at boot — can happen after kernel updates due to USB bus timing on the Pi B+.

Solution: The `wlan-driver-check` service (section 11) detects and recovers this automatically within ~60 seconds of boot. Check its status with: `journalctl -b 0 | grep wlan-driver-check`

gmap.html shows 'Problem fetching data from dump1090'

Cause: `fr24feed` is spawning its own `dump1090` instance (without `--net`) instead of using the standalone one.

Solution: Ensure `fr24feed.ini` uses `receiver="beast-tcp"` and `host="127.0.0.1:30005"`, and that the standalone `dump1090-mutability` service is running.

dump1090-mutability shows 'disabled' and won't start

Cause: `START_DUMP1090` is set to "no" in `/etc/default/dump1090-mutability`.

Solution: `sudo sed -i 's/START_DUMP1090="no"/START_DUMP1090="yes"/' /etc/default/dump1090-mutability`

fr24feed shows 'Connection refused' connecting to dump1090

Cause: `fr24feed` started before `dump1090-mutability` was ready, or incorrect host/port in `fr24feed.ini`.

Solution: Ensure `host="127.0.0.1:30005"` (port included in host string) in `fr24feed.ini`, then restart `fr24feed` after `dump1090` is running.

14. Important Files to Back Up

File	Purpose
<code>/etc/fr24feed.ini</code>	FR24 feeder config including your <code>fr24key</code>

/etc/default/dump1090-mutability	dump1090 startup configuration
/etc/modprobe.d/blacklist-rtl.conf	RTL-SDR kernel module blacklist
/etc/NetworkManager/system-connections/	WiFi connection profiles
/etc/NetworkManager/conf.d/wifi-powersave-off.conf	WiFi power saving config
/etc/systemd/journald.conf	Persistent journal configuration
/usr/local/bin/wifi-watchdog.sh	WiFi watchdog script
/usr/local/bin/wlan-driver-check.sh	WiFi driver recovery script
/etc/systemd/system/wlan-driver-check.service	WiFi driver recovery systemd service

15. Quick Reference Commands

Check all service statuses

```
sudo systemctl status fr24feed dump1090-mutability lighttpd wlan-driver-check
```

View fr24feed live log

```
sudo journalctl -fu fr24feed
```

Verify only one dump1090 instance

```
ps aux | grep dump1090
```

Check listening ports

```
sudo ss -tlnp | grep -E 'dump1090|lighttpd'
```

Check WiFi status

```
ip a s wlan0 && nmcli connection show --active
```

Restart all services

```
sudo systemctl restart dump1090-mutability && sudo systemctl restart fr24feed && sudo systemctl restart lighttpd
```

Check journal from previous boot

```
journalctl --list-boots
journalctl -b -1 -u NetworkManager
journalctl -b -1 | grep -E 'wifi-watchdog|wlan-driver-check'
```

Check journal disk usage

```
journalctl --disk-usage
```

Manually trigger driver recovery check

```
sudo /usr/local/bin/wlan-driver-check.sh  
journalctl | grep wlan-driver-check | tail -5
```